

Julie Ha

Contact Information

Website: <https://hajulie.com>

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Education

Boston University Computing and Data Sciences Boston, MA

Ph.D. Candidate

Advisor: Mayank Varia

Research Interests: applied cryptography, computer science and law, usable security and privacy

Boston University Boston, MA

B.A. in Computer Science with Honors

Recipient of Computer Science Award for Academic Excellence

Senior Thesis: Oblivious Searchable Encryption for Irises

September 2018 - May 2022

Publications

- [1] R. Canetti, J. Ha, G. Kaptchuk
Analyzing Cryptography in Context: A Cryptography-Native Approach to Threat Modeling
In Submission to Usenix 2026
 - Contribute a framework for analyzing sociotechnical goals of cryptographic deployments
- [2] J. Ha, C. Cachet, L. Demarest, S. Ahmad, B. Fuller
Private Eyes: Zero-Leakage Iris Searchable Encryption.
ACM Conference on Data and Application Security and Privacy (CODASPY)
<https://eprint.iacr.org/2023/736>
 - Contribute the first *zero-leakage* biometric database using locality sensitive hashes (LSHs) and oblivious maps
 - Implemented Private Eyes in Python and tested implementation on real, synthetic, and random irises
- [3] Y. Gvili, J. Ha, S. Scheffler, M. Varia, Z. Yang, and X. Zhang
TurboIKOS: Improved Non-interactive Zero Knowledge and Post-Quantum Signatures.
International Conference on Applied Cryptography and Network Security (ACNS) 2021
<https://eprint.iacr.org/2021/478>
 - Contribute a new zero-knowledge argument proof in the MPC-in-the-head paradigm (MPCitH) that retains all benefits of MPCitH while providing concretely smaller proof sizes
 - Implemented TurboIKOS in Python

Works in Progress

- [1] Measuring Content Transformation for Fair Use
 - Contribute a formal technical definition of a fair-use property using elements of Kolmogorov complexity
- [2] Digital Security and Privacy in Immigration Rights Organization
 - Interview personnel at immigration-rights organizations to identify digital security and privacy concerns related to their use of existing digital tools
- [3] Private Borrowing and Calendar Appointment System
 - Building a cryptographically secure system for borrowing systems, such as libraries and bike shares, and calendar appointment systems, that prioritizes user privacy

Awards

CDS Teaching Excellence Award (May 2026)

Hariri Institute Community Recognition Awards - Graduate Student Excellence Award (June 2023)

Teaching Experience

Fall 2025: Graduate TA for DS653: Crypto for DS

Spring 2025: Graduate TA for DS593: Privacy in Practice

Spring 2023: Graduate TA for CS558: Network Security

Spring 2022: Grader for CS538: Fundamentals of Cryptography, Grader for CS558: Network Security

Fall 2021: Undergraduate Teaching Assistant for CS357: Introduction to Information Security

Talks

Building a Guidebook for Collaborations between Cryptographers and Social Good Organizations

Annual Boston Security Usability Research Day (ABSURD) (February 2026)

Private Eyes: Zero-Leakage Iris Searchable Encryption

ACM Conference on Data and Application Security and Privacy (June 2025)

University of California, Berkeley Security Seminar (May 2025)

TurboIKOS: Improved Non-interactive Zero Knowledge and Post-Quantum Signatures

University of Connecticut Security Seminar (July 2021)

International Conference for Applied Cryptography and Network Security (June 2021)

Posters

Cryptography in Context: The Case Study of Apple's CSAM Scanning Proposal

New England Security Day 2024

Annual Boston Security Usability Research Day (ABSURD) 2024

External Reviewer

Privacy Enhancing Technologies Symposium (PETS) 2026

USENIX Security 2025

CCS 2024

USENIX Security 2023

CCS 2023

USENIX Security 2022

Crypto 2022

Outreach

CodeBreakers Boston, MA (Remote)

Coordinator

- Instructed cryptography portion of a summer program for high school sophomores and juniors to help create relationships in STEM and introduce topics in computer science
- Created activities related to computer science to facilitate engagement remotely

June 2020 - August 2020

References

Available upon request